Rawja Alam

Dhaka, Bangladesh rawja.alam@gmail.com Scholar ORCID Website

Academic Credentials

Military Institute of Science and Technology *Feb 2020 – April 2024*
Bachelor of Science in Civil Engineering (Major: Water, Minor: Structure)

- GPA: 3.47/4.0
- **Coursework:** River Engineering, Hydraulic Structures, Hydraulic Structures Design Sessional, GIS and Remote sensing, Engineering Hydrology and Irrigation Engineering, Design of Water Supply, Sanitation, Fundamentals of Transportation Engineering, Highway Materials, Pavement Design, Railway Engineering, Traffic Engineering, Finite Element Method, Design of Steel-Concrete Composite Structure

Standardized Test

International English Language Testing System (IELTS Academic) *22 Oct, 2026*
Overall Band Score: 8.0
L: 9.0 R: 8.5 W: 7.0 S: 7.0

Graduate Record Examination (GRE) *13 Oct, 2026*
Total: 310 Quantitative: 160 Verbal: 150 Analytical Writing: 4.0

Work Experience

Research Assistant *Dhaka, Bangladesh*
Bangladesh University of Engineering and Technology (BUET) *Nov 2025 - Present*

- Working on Web-based Agricultural Information System for Bangladesh (WebAIS) Project
- Developing a DDIEM-inspired, grid-based water and salt balance model in MATLAB for Polder 30
- Integrating hydro-climatic, land-use, cropping, flow-routing, and salinity data for daily soil salinity prediction

Internship *Dhaka, Bangladesh*
Ministry of Health and Family Welfare (Health Engineering Department) *February 2023*

- Monitored design details, site construction process, and cost estimation of work
- Observed the specific methods in sheet pile construction, slab and shutter design, and site safety measures and highlighted disparities between hospital and general construction processes
- Prepared presentations and documents for the project and BOQ

Journal Publications

Reimagining Transportation Infrastructure for the Autonomous Era: A Comprehensive Review *June 2026*
Saki Rezwana, ***Rawja Alam***, Devin Rhoads, Nicholas Lownes
DOI: 10.3390/app16115704

Assessing Riverbank Transformations of the Major Rivers in Bangladesh: A Remote Sensing and GIS based Study of Erosion and Accretion Patterns from 1988-2023 *April 2025*
Quazi Afra Nawar, ***Rawja Alam***, Jumana Akhter, G. M. Jahid Hasan
DOI: 10.1080/01490419.2025.2494695

Conference Publications & Abstracts

Spur Protection Failure Due to Bend-Scale Instability at the Panka-Narayanpur Reach, Ganges-Padma River *(Accepted)* *2026 AGU Annual Meeting*
Sameer Mahmud Khan, ***Rawja Alam***, Md. Munsur Rahman

Multi-Sectoral Impact Assessment and Flood Risk Zoning of Eastern Bangladesh for the August 2024 Flood *(Accepted)* — *ICCEI 2025*

Naimul Hasan, Nusaiba Nueri Nasir, Mairukh Wardi, ***Rawja Alam***, Chandan Kumar Sarker, Rakibuzzaman Shafi Siam, Kh. Al Awual Arabi, Aditya Azim

Redefining Value Creation: Assessing the Economic Impact of Digital Twin Across Industries *(Accepted)* — *IEOM 2024 World Congress, Detroit, Michigan*

Md Tarique Hasan Khan, Saki Rezwana, ***Rawja Alam***

Research Work

ASSESSING THE BANKLINE DYNAMICS OF THE MAJOR RIVERS OF BANGLADESH USING GIS AND DSAS TECHNIQUES (Thesis)

This study investigates the dynamics of riverbank erosion and accretion in the major rivers of Bangladesh using remote sensing and GIS techniques. Landsat satellite imagery was collected and processed, and NDVI (Normalized Difference Vegetation Index) was applied to evaluate vegetation cover and environmental conditions along riverbanks. River boundaries were digitized to generate GIS-based spatial datasets for hydrological analysis. The Digital Shoreline Analysis System (DSAS) was used to calculate erosion and accretion rates across different time scales, enabling the quantification and assessment of riverbank changes under both short-term and long-term conditions.

MOTORCYCLE OWNERSHIP AND USAGE PATTERNS AMONG INCOME GROUPS IN DHAKA: IMPLICATIONS FOR PUBLIC TRANSPORTATION SYSTEMS (Ongoing)

This ongoing study explores the distribution and utilization of motorcycles across different income groups in Dhaka. The project aims to understand how rising motorcycle ownership influences travel behavior and affects the performance and planning of public transportation systems.

REVIEW ON EARTH-CORE ROCKFILL DAMS (Ongoing)

I am co-authoring a review paper focusing on the structural stability and environmental challenges associated with earth-core rockfill dams. My contributions include collecting and organizing published research data, conducting a scientometric analysis to assess global research trends, and critically evaluating design structures, performance limitations, and resilience under hydrological and seismic stresses.

GLOBAL DAM MAPPING USING GIS (Ongoing)

In this ongoing project, I am developing an interactive GIS-based global dam database. This work involves collecting dam location and attribute data from international sources and satellite imagery, processing spatial datasets, and creating visualizations to highlight dam distribution and characteristics.

Technical Strengths

Software Skills: AutoCAD, SAP2000, ETABS, HECRAS, ArcGIS Pro, ArcGIS Desktop, QGIS

Programming Languages: C/C++, Python, MATLAB

Others: Latex, Unreal Engine, Adobe Illustrator, Microsoft Excel

Achievement and Extra Curricular Activities

University Merit Scholarship in MIST (Level-2)

- In recognition of academic excellence in Undergraduate studies

Elected as Class Representative, Department of Civil Engineering, MIST

- By votes of students and recommendation of faculty members (2020)

Certifications

Reviewer Certificate, *Catena* — *March-November 2025*

- Reviewed 7 manuscripts on climate-driven vegetation changes and sediment-discharge modeling.

Global Environmental Management, *Coursera* — *June 27, 2020*
Technical University of Denmark (DTU)

Introduction to Engineering Mechanics, *Coursera* — *June 21, 2020*
Georgia Institute of Technology

C for Everyone, Part 1: Programming Fundamentals, *Coursera* — *June 13, 2020*
University of California, Santa Cruz

References

Dr. G M Jahid Hasan
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Military Institute of Science and Technology (MIST)
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